

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

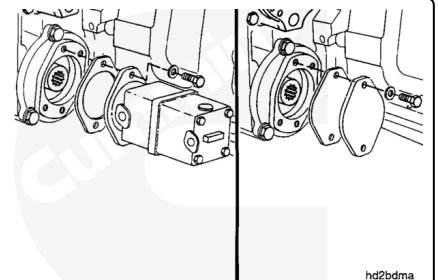
- Barring tool, Part Number 3824270
- Capscrew, Part Number 3335003 or 3018671, or equivalent - Quantity two
- Gear housing bearing install/remove tool, Part Number 3823774
- Dial indicator assembly, Part Number 3376050
- Extension, Part Number ST-537-4
- Magnetic base indicator holder, Part Number 3377399

Additional Service Items

- Slide hammer
- Lubriplate™ 105, or equivalent

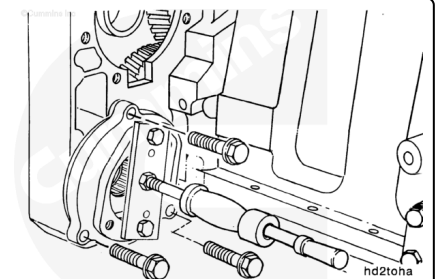
Remove

Remove the two mounting capscrews and hydraulic pump, if equipped, or the cover plate.



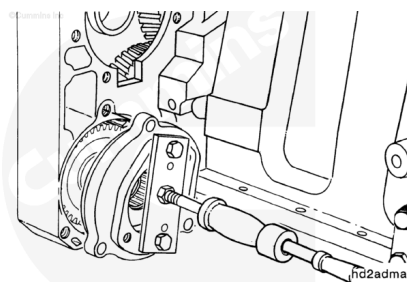
Remove the three pump drive mounting capscrews.

Install a slide hammer and engine barring tool, Part Number 3824270, to the hydraulic drive adapter. Use two M10-1.50 x 30 capscrews, Part Number 3335003, for Society of Automotive Engineers (SAE) A drives, or two M12-1.75 x 40 capscrews, Part Number 3018671, for SAE B drives.

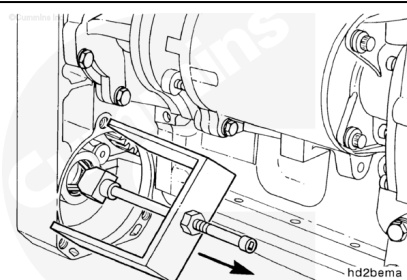


Do not allow the hydraulic drive gear to slide out of the adapter or the gear housing during removal. Damage to the gear will result.

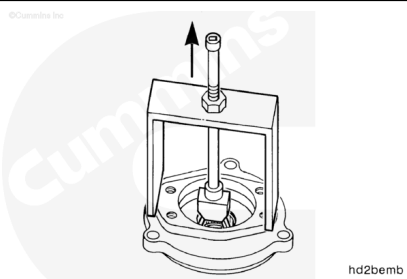
Remove the hydraulic drive adapter, o-ring, and hydraulic drive gear.



Use gear housing bearing install/remove tool, Part Number 3823774, to remove the needle bearing from the gear housing.

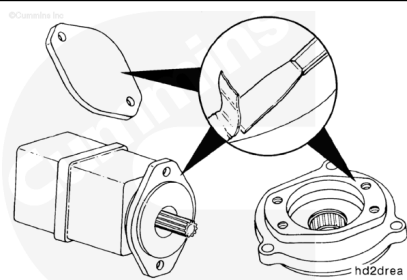


Remove the needle bearing from the hydraulic pump adapter.



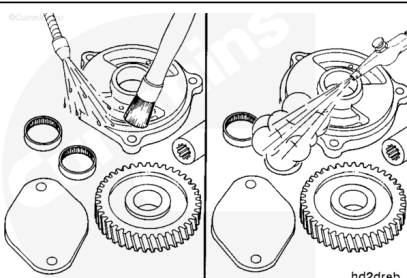
Clean and Inspect for Reuse

Remove the gasket material from the hydraulic pump adapter and hydraulic pump, if equipped, or cover plate surface.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

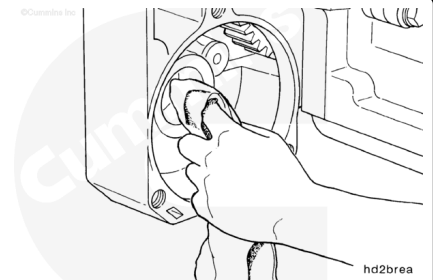


⚠ WARNING ⚠

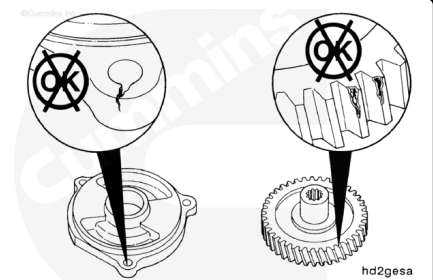
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the hydraulic pump adapter, cover plate, gear, and needle bearings. Use solvent. Dry with compressed air.

Clean the needle bearing bore in the gear housing. Use solvent. Dry with a clean cloth.

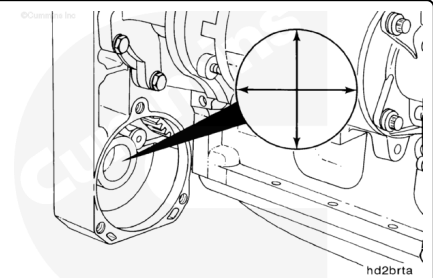


Inspect the components for damage.



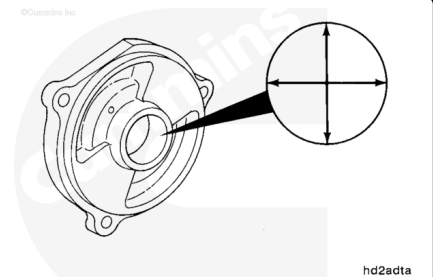
Measure the inside diameter of the needle bearing bore in the gear housing.

Needle Bearing Bore Inside Diameter (Gear Housing)		
mm		in
41.967	MIN	1.6522
41.992	MAX	1.6532



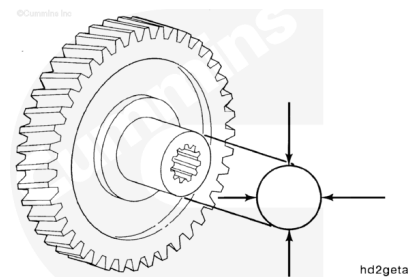
Measure the inside diameter of the needle bearing bore in the hydraulic pump drive adapter.

Needle Bearing Bore Inside Diameter (Hydraulic Pump Drive Adapter)		
mm		in
41.967	MIN	1.6522
41.992	MAX	1.6532



Measure the hydraulic drive shaft outside diameter.

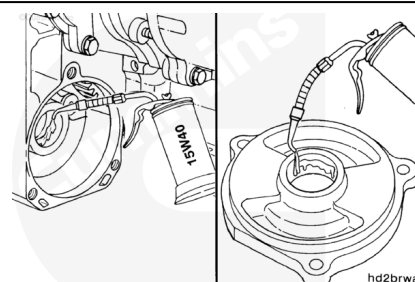
Hydraulic Drive Shaft Outside Diameter		
mm		in
34.984	MIN	1.3773
35.000	MAX	1.3780



Install

Use clean 15W-40 oil to lubricate the outside diameter of the needle bearings.

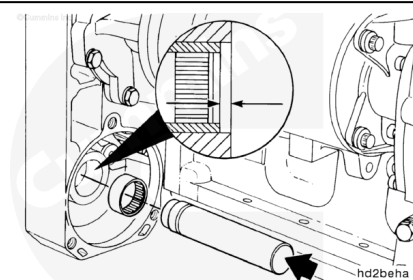
Use clean 15W-40 oil to lubricate the needle bearing bores in the gear housing and hydraulic pump drive adapter.



Use bearing installation tool, Part Number 3823776, included in gear housing bearing install/remove tool, Part Number 3823774, to install the needle bearings in the gear housing.

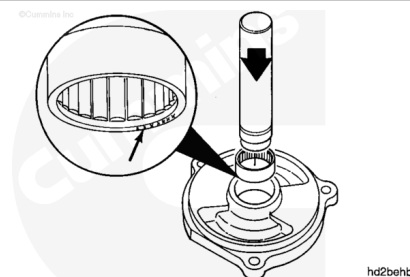
Tap the bearing gently until it comes in contact with the shoulder in the housing.

The bearing **must** be 0.25 to 0.76 mm [0.010 to 0.030 in] past the outside edge of the gear housing bore surface.



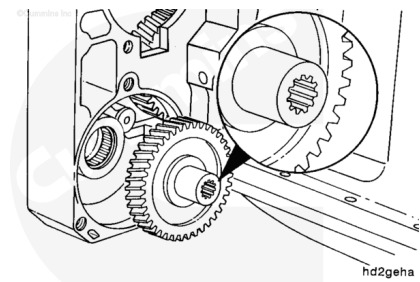
Install the needle bearing in the hydraulic pump drive adapter from the inside end (the end that will be closest to the hydraulic drive gear).

The bearing **must** be centered between the front and rear surfaces.



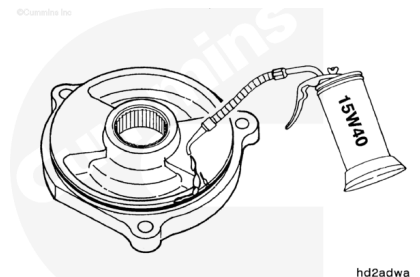
Use Lubriplate™ 105, or equivalent, to lubricate the needle bearing in the gear housing and the outside of the hydraulic driveshaft.

Install the shaft and gear assembly so the internal splines of the shaft are facing away from the gear housing and toward the rear of the engine.



Install a new o-ring on the hydraulic drive adapter. Lubricate the o-ring with clean 15W-40 oil.

Lubricate the hydraulic drive adapter needle bearing and bore. Use Lubriplate™ 105, or equivalent.



Install the hydraulic drive adapter over the shaft and gear assembly.

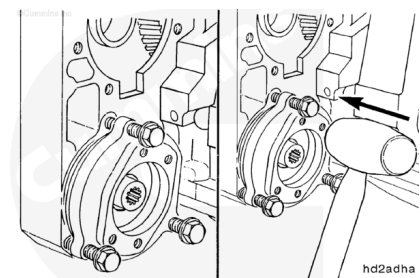
Start the three mounting capscrews by hand to align the hydraulic drive adapter with the gear housing.

Using a rubber hammer, tap around the outside diameter of the adapter to install the adapter.

Note : Use care not to damage the o-ring during installation.

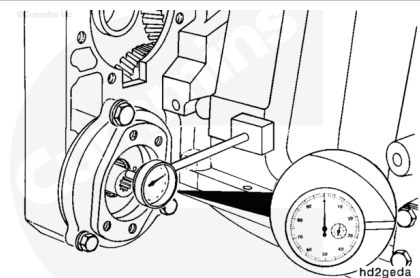
Tighten the mounting capscrews.

Torque Value: 47 n•m [35 ft-lb]



Measure driveshaft end clearance.

Use dial indicator assembly, Part Number 3376050, with extension, Part Number ST-537-4, and magnetic base indicator holder, Part Number 3377399.



Hydraulic Pump Driveshaft End Clearance		
mm		in
0.076	MIN	0.003
0.635	MAX	0.025

Use a new gasket. Install the hydraulic pump, if equipped, or the cover plate.

Install the two mounting capscrews. Tighten.

Torque Value:

Grade 8.8/9.8 M10 capscrew 36 n•m [27 ft-lb]

Torque Value:

Grade 10.9 M10 capscrew 51 n•m [38 ft-lb]

Torque Value:

Grade 8.8/9.8 M12 capscrew 62 n•m [46 ft-lb]

Torque Value:

Grade 10.9 M12 capscrew 88 n•m [65 ft-lb]

